

The application is composed of different components as well as packages for different tasks. Each task package has its own target functionality. What makes the design of this overall application stand out is that it does not have duplicate code or blocks of code across different task packages that share the same functionality. It promotes code reusability, especially from Task 3 to Task 8, where target functionalities are connected from top to bottom, as each achieves its target function without wasted code.

The application's use of utilities such as BullsCounter, CowsCounter, OutputFormatter, and ValidDigitGenerator enables the application to have centralized validation and result services throughout development. The use of Singletons also allows the application to avoid excessive dependency injection, as it enables the use of a single entity for configuration and secret codes across different files within a single runtime.

The initial design of the application was to have different task files independent from each other, but this was changed because possible changes in the game mechanics or new features would require the developer to check each task file one by one for potential updates. Through various test cases, the performance and correctness of the application were ensured.

The revised sequence diagram was expanded to show the complete flow of the Bulls and Cows game instead of only the guessing part. It now includes the game setup, where the player configures the settings, chooses whether to use manual input or a file for guesses, and selects the AI difficulty. It also adds the step where both the player and the computer set their secret codes before the game starts. During the game, the diagram now shows both the player's and the computer's turns, along with the checking of bulls and cows after every guess. It also includes more validation steps, such as checking the game configuration, secret code, and file path, not just the guess format. Finally, the updated diagram shows all possible game endings, including player win, computer win, or draw, and adds the option to save the game log to a text file after the game ends.